

Chapter 23

Grammatical tone in African languages

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1 Introduction

This chapter will define, describe and illustrate grammatical tone in a number of African languages. Admittedly, this is a topic that strikes fear in the heart of many researchers. And not without cause. Hyman (2011) noted that “since some tonal phenomena have no segmental or stress analogues, anyone who is interested in the outer limits of what is possible in phonology would thus be well-served to understand how tone systems work”, and grammatical tone is exemplary of such ‘outer limits’. Along similar lines, Kisseberth & Odden (2003: 70) in their discussion on Bantu remarked that some of its tone systems reveal “patterns of extraordinary complexity”, of which grammatical tone is part-and-parcel. In this paper we hope to make at least some progress in demystifying grammatical tone, by carefully defining what it is and providing enough examples so that current African language researchers become more confident in identifying and describing such patterns in the languages they work on.

In this first section, we discuss the difference between lexical and grammatical tone. We also describe the kinds of things that grammatical tone is used for as well as the range of ways in which grammatical tone can be employed. Grammatical tone will then be illustrated in various African languages, organized based on

Greenbergian language phylum: Niger-Congo (Section 2), Nilo-Saharan (Section 3), Afro-Asiatic (Section 4), and finally Khoisan (Section 5).

We note here that in a paper of this limited length, it is obviously impossible to document every claim of a grammatical tone operating in some African language. We have therefore endeavored to include the most interesting and instructive cases which illustrate the effects of grammatical tone. In some language families there are relatively few reliable descriptions, while in others there is an abundance of them. In the latter groups, while ultimately choosing which specific languages to highlight is by definition subjective, we're confident that the range and depth of the examples presented will accomplish the goal of familiarizing the reader with the many ways in which grammatical tone is used in African languages.

Throughout this chapter, we shall largely employ diacritics on the vowels to represent vowels, following the International Phonetic Alphabet conventions (i.e. à = super-low/sL, à = low/L, ā = mid/M, á = high/H, and ã = super-high/sH, and various contours formed from these). We note overtly when we deviate from this transcription practice. In cases where we provide no tone diacritic, this should be interpreted as a tonally unmarked syllable (i.e. toneless), which gets its phonetic pitch value by default (usually, surfacing as low).

1.1 Lexical vs. grammatical tone

We can separate tonal function into two broad categories: lexical versus grammatical. Lexical tone is part of the underlying form of a morpheme, on par with nasality, vowel height, sonority and other contrastive phonological features. This is most easily illustrated with minimal pairs, such as the infinitival forms in the Bantu language Setswana (ISO 639-3 code [tsn] – Botswana), as provided in their phonetic forms in Table 1.

Table 1: Setswana tonal minimal pairs on verbs (Jones et al. 2018)

Low-toned roots		High-toned roots	
[gò-dùm-à]	‘to roar’	[gò-dúm-à]	‘to spray with insecticide’
[gò-sèlw-à]	‘to pick up’	[gò-sélw-à]	‘to oversleep’
[gò-bàk-à]	‘to praise in song’	[gò-bák-à]	‘to bake bread’

Each member of the three pairs of forms above are segmentally identical – the only thing that distinguishes them is tone. The infinitival prefix *gò-* always surfaces as low, as does the Bantu final vowel *-à*. The tonal difference is situated

solely within the root. Given the fact that the roots are segmentally identical, it is impossible to predict the tone on the basis of any phonological aspect of the root, and we therefore may conclude that tone is contrastive on verb roots in this language. Looking deeper at the language, one must decide how to interpret the phonological status of this two-way contrast, i.e. as H vs. L (e.g. /sélw-/ ‘oversleep’ vs. /sèlw-/ ‘pick up’), as H vs. toneless ($\emptyset$) (e.g. /sélw-/ vs. /selw-/), or perhaps as $\emptyset$ vs. L (e.g. /selw-/ vs. /sèlw-/). See **chapters/floating_tones** of this volume for discussion on positing toneless syllables.

Moreover, if one has only the data in Table 1, it might not be clear whether the High tone in Setswana should be pre-associated to its vowel or merely be sponsored by it and come in ‘floating’. We maintain that this is a separate analytic issue. What is clear is that a High tone must be part of the root meaning ‘oversleep’, and thus it is lexical. Depending on the tonal patterns of these forms in their simple and derived instantiations, the lexical High might be argued to be pre-linked or floating. See **chapters/floating_tones** of this volume for an overview of ‘floating tones’.

The other thing to emphasize here is that lexical tones can potentially be present in any type of root, as well as in affixes. Lexical tone is a property not just of verbs (as in Table 1), but can be a property of any part of speech, including nouns, adjectives, adverbs, pronouns, clitics, prefixes, suffixes, and so on. A comparable example of tonal minimal pairs in Setswana nouns is in Table 2 (note the noun class prefix before the noun root in the latter two rows).

Table 2: Setswana tonal minimal pairs on nouns (Jones et al. 2018)

Low-toned roots		High-toned roots	
[thàkà]	‘pupil (of eye)’	[thákà]	‘friend, peer’
[mà-fùlò]	‘pastures’	[mà-fúlò]	‘foam, froth’
[mà-bòkò]	‘praise poems’	[mà-bòkó]	‘brains’

That lexical tone is not limited to roots is demonstrated below, with a three-way minimal pair of prepositions from Aghem ([agq] – Grassfields: Cameroon) in Table 3 and a two-way minimal pair of derivational suffixes in Herero ([her] – Bantu: Namibia) in Table 4. (Note the presence of a downstep¹ in Table 3 which indicates the lowering of a high tone’s register, but not as low as a true Low tone.)

In these tables, the lexical tone is part of the underlying representation of an individual morpheme and appears on that morpheme. It does not appear in all comparable parts of speech (e.g. all nouns, or all suffixes), and the tone is not

Table 3: Aghem prepositions (Hyman 2010)

High-toned locative	á ‘in’	á ‘ndúghó	‘in the house’
Falling-toned reciprocal	â ‘to’	â bvâ ‘tó	‘to the dogs’
Low-toned comitative	à ‘with’	à bvâ ‘tó	‘with the dogs’

Table 4: Herero suffixes (Hyman 2018, citing Wilhelm Möhlig, p.c.)

Toneless stative suffix	-ak	zúv-ak-a	‘be heard’
High-toned neutro-passive suffix	-ák	zúv-ák-a	‘be hearable’

extractable and does not mean anything on its own (i.e. in Aghem, Low tone does not combine with other prepositions with a meaning ‘comitative’). This contrasts with the subject of this chapter, namely grammatical tone. Here we adopt a definition of grammatical tone as a productive tone alternation triggered by a particular grammatical context (rather than purely a phonological context), which functions to express the meaning of that grammatical context.

Let us briefly consider the parts of this definition. First, the alternation must be productive – that is, it cannot be exhibited by only a few morphemes, but rather targets an unbounded class. If a grammatical tone expressing Remote Future affects one verb stem, it should affect all of them (all else being equal). Second, the tone alternation cannot be due to a language-wide rule which applies whenever a certain phonological contexts is met, e.g. a predictable phonological rule inserting a H tone between two any two L tones, regardless of morphological context. Such a process is more akin to segmental epenthesis rule, and likewise grammatical tone is more akin to segmental affixation. Third, grammatical tone signals meaning – that is, it is part of the obligatory morphological expression of a linguistic category. One can forego grammatical tone as much as one can forego deriving an agentive noun or inflecting a finite verb: almost never, and only under highly specific circumstances.

1.2 Types of grammatical tone

How exactly is grammatical tone added to or implemented on the base when it is found? The first important point to make is that very often it’s not just the presence of a grammatical tone (or sequence of tones) on its own which expresses some grammatical meaning. In situations where that is the case, we will refer

to that as ‘autonomous grammatical tone’ (also referred to in the literature as ‘tonal morphs’ or ‘morphemes’). But often grammatical tone is one of several co-exponents of some grammatical expression, and here we refer to it as ‘non-autonomous grammatical tone’. For instance, some languages might express the Potential mood by a combination of a segmental verbal prefix and a grammatical tone suffix, others by tone alone.

A partial taxonomy of grammatical tone patterns includes several different types, such as tone addition, subtraction, replacement, shifting, exchanging, and spreading. We provide a very brief description of these here, as they will be illustrated more fully in Sections 2-5 to come.

Tone Addition involves the insertion of a grammatical tone on a specific Tone Bearing Unit (TBU). This is the case in Makonde ([kde] – Tanzania) in Table 5, where various inflectional contexts add a grammatical tone pattern onto the penultimate TBU, either a Low ① in the hypothetical (HYP), a High ② in the consecutive (CONS), a ①② in the future (FUT), or a ①②① in the remote past (RPST).

Table 5: Makonde additive grammatical tone (Manus 2014: 267–268)

Low-toned roots		High-toned roots	
HYP	káléká ngu-①-takatukil-e	→	... ngu-takatuk[ìì]le ‘if I stood up’
CONS	ni-ka②-takatukil-a	→	ni-ka-takatuk[íí]la ‘and I stood up’
FUT	ni-nda①②-takatukil-a	→	ni-nda-takatuk[íí]la ‘I will stand up’
RPST	ní-ndí①②①-takatukil-a	→	ní-ndí-takatuk[íì]la ‘I stood up’

Note that in Table 5, we denote the grammatical tone in a circle, a convention which we will continue this point forward. The circled tone is generally reserved for ‘floating tones’ (introduced above), which may be intuitive for some grammatical tone patterns but much less so for others. We use the circle convention for convenience in this paper, and acknowledge that other formalizations/representations are possible for many of the grammatical tone patterns we highlight below.

Moving on, when some grammatical expression involves the deletion of some lexical tone this is a case of Tone Subtraction. For instance, the addition of some segmental suffix might delete all of the tones within the root to which it attaches, leaving other productive tonal processes to value the TBUs which lost their tonal specification in this process (such as late default fill-in rules). Consider the Ijoid language Nkoroo ([nks] – Nigeria). In Noun-Noun compounds, the underlying tones of the second noun (the head noun) delete, as represented below in the

intermediate forms in Table 6. How the tones on the second noun ultimately surface depends on the first noun’s underlying tones. If the first noun ends in H then the second noun surfaces [HL], but if the first noun ends in L then the second surfaces as [LL]. This can straightforwardly be understood as tone spreading to the toneless second noun, plus default low on all non-initial vowels.

Table 6: Nkoroo subtractive grammatical tone in Noun-Noun compounds (Akinlabi et al. 2003)

Noun 1		Noun 2		Intermediate		Surface
bís	‘leg’	fúrò	‘belly’	→	bís furo	→ [bís fúrò] ‘thigh’
ánáná	‘sheep’	tòkú	‘child’	→	ánáná tòku	→ [ánáná tókù] ‘lamb’
ḡḡḡkò	‘chicken’	tòkú	‘child’	→	ḡḡḡkò tòku	→ [ḡḡḡkò tókù] ‘chick’
wàrì	‘house’	tòkú	‘child’	→	wàrì tòku	→ [wàrì tókù] ‘servant’

Relatedly, Tonal Replacement occurs when existing lexical tones are not simply deleted, but systematically replaced by some grammatical melody. Unlike in Nkoroo, in some languages where underlying tones are deleted, a fixed grammatical tone provides the replacement, regardless of the underlying tone of triggering environment. In a related Ijoid language Kalabari ([ijn] – Nigeria), the tones of the noun are deleted and replaced by a uniform ①② pattern in the context of a proximate demonstrative ‘this’, shown in Table 7.

Table 7: Kalabari replacive ①② tone (Harry & Hyman 2014)

HH	námá	‘animal’	→	mí námá	‘this animal’
LL	pùlò	‘oil’	→	mí pùlò	‘this oil’
HL	bélè	‘light’	→	mí bèlé	‘this light’
LH	gàrí	‘garri flour’	→	mí gàrí	‘this garri flour’
H’H	ḡárá	‘hand’	→	mí ḡàrá	‘this hand’

Next, Tone Shifting involves the shifting of an underlying (lexical) tone to a different TBU within the same morpheme. For instance, a particular suffix might induce the shift of a root-initial tone to some other TBU. Crucially, it would

need to be demonstrated that this shift was not a general rule (which would be triggered by any suffix) nor can it be phonologically predictable (e.g. occurring when the tone was followed by some number of TBUs). Tone Shifting is found in the Bantu language Tachoni ([lts] – Kenya), where in the imperative, toneless stems remain toneless but stems with initial high tone shift this high to the penult (which sometimes appears additionally one TBU to the left). This is shown in Table 8.

Table 8: Tachoni shifting tone in imperative (Odden 2009)

Toneless stems			H-toned stems		
chiingila	→	[chiingila] ‘carry for!’	káraanga	→	[karáánga] ‘fry!’
beechakala	→	[beechakala] ‘belch!’	bótooxanila	→	[botooxáníla] ‘go around for!’

The last kind of grammatical tone and related to the last one is Exchanging Tone. One such case would be a stem with an underlying H losing it (like subtractive tone) but a stem without any underlying tone gaining a H (like additive tone). Exchanging tone can also be seen in Tachoni, in its negative imperative. Simplifying a bit, the data in Table show that toneless stems are assigned a H which appears after the prefix *xa-*, whereas H-toned stems surface toneless with neither underlying tone nor any grammatical tone (what Odden calls ‘mutual annihilation’).

Table 9: Tachoni exchanging tone in negative imperative

Toneless stems			H-toned stems		
bala	→	[o-xa-bála] ‘don’t count!’	lúma	→	[o-xa-luma] ‘don’t bite!’
beechakala	→	[o-xa-bééchakala] ‘don’t belch!’	káraanga	→	[o-xa-karaanga] ‘don’t fry!’

Other grammatical tone patterns exist as well, such as grammatically-induced spreading operations (where no general spreading rule would account for the pattern), or other more phonologically natural processes (e.g. assimilation, dissimilation, raising, lowering, plateauing, etc.). We point out, however, that natural

tone rules which have highly specific prosodic or (morpho)syntactic conditioning are frequently excluded from grammatical tone discussion. For instance, if a language has a rule which deletes a High tone before another High tone, but only within a stem, or alternatively within a phonological phrase. We do not classify such cases as grammatical tone, as the pattern resulting from this change signals no change in meaning.

1.3 What is grammatical tone used for?

Given the definition and range of grammatical tone patterns in the previous subsection, what can grammatical tone be used for? One striking way to respond to this would be: any inflectional or derivational process which a segmental morpheme can express, tone too can express! Perhaps Hyman (2011) said this best: “I conclude that tone can do everything that segmental and metrical phonology can do, but that the reverse is not true”. This appears to hold, and no strict gaps are known where an inflectional or derivational process can only be expressed by segments and never tone. However, it may very well be the case that grammatical tone is used to express some inflectional and derivational properties more than others, e.g. verbs commonly inflecting for tense/aspect/mood via grammatical tone, or nouns commonly marked by grammatical tone in compounds/possessives/associative constructions. At the same time, we also find grammatical tone commonly used syntactically, for example, in syntactic agreement between a head and modifier or complement. In Sections 2-5 below, we will describe specific cases of grammatical tone in a number of African languages which illustrate how it is used.

2 Niger-Congo

The Niger-Congo phylum is one of the largest in the world, and even if one excludes its more controversial branches (e.g. Ijoid, Mande, Kordofanian, among a few others), it still constitutes one the largest families in the world. Grammatical tone is rampant in Niger-Congo, and it is hard to find languages within which grammatical tone does not play a major role. We shall go through several major sub-branches and discuss common grammatical tone patterns (though for considerations of space, we simply cannot explore each and every sub-branch, and leave out discussion of Adamawa, Ubangi, Gbaya, Senoufo, Dogon, and various ‘Atlantic’ families).

2.1 Benue-Congo

Perhaps the most famous sub-branch of Niger-Congo is Benue-Congo, no doubt due to the presence of its expansive sub-sub-branch Bantu. We have seen several cases of Bantu already, such as Makonde in Table 5 above where a particular grammatical tone pattern is part of the expression of verb inflection. Such patterns are omnipresent in Bantu, and can form some of the most complex tonal patterns known to linguistics.

Consider the Bantu language Kuria ([kuj] – Kenya), which has become well-known for its ‘counting effects’ in grammatical tone assignment (Paster 2019). Table 10 provides a mini-paradigm involving the prefix *to-* 1p ‘we’ with the verb root *kirigiit-* ‘scrub’ in various inflected contexts. Depending on the grammatical context, a grammatical ⑥ tone is systematically assigned to the first, second, third, or fourth mora of the stem (denoted in curly brackets), and even to the first and fourth simultaneously. Note that the forms show the effects of general tone spreading rules. Where the grammatical tone is assigned originally is the underlined TBU of the stem.

Table 10: Kuria counting effects (Mwita 2008)

⑥ ₁	Habitual past	n-to-ogá-{ <u>k</u> irigiít-íre}	‘we used to scrub (then)’
⑥ ₂	Past prog. (foc.)	n-to-oga-{k <u>i</u> rigiít-íre}	‘(indeed) we have scrubbed’
⑥ ₃	Remote fut. (foc.)	n-to-re-{k <u>i</u> rigiít-a}	‘we will scrub (then)’
⑥ ₄	Inceptive	to-ra-{k <u>i</u> rigiít-a}	‘we are about to scrub’
⑥ ₁₊₄	Narrative past	to-gá-{ <u>k</u> irigiít-a}	‘(and) we scrubbed’

If we assume that grammatical tone generation and docking both take place in the morphology, we might expect that these processes should be confined to the morphological word. While this is true for many languages, there is a variety of Kuria which convincingly shows that grammatical tone assignment must be done at the level of the phrase instead of the word. Marlo et al. (2014) show that in the Buguumbe dialect of Kuria, speakers realize some verbal grammatical High tones within the *following* word. For example, the Inceptive targets the fourth mora of the stem, and when the stem has four morae then the grammatical ⑥ docks onto that stem-final fourth mora (as in the first row of Table 11). But when the stem is shorter than four morae, the ⑥ docks within the following word, on whatever is the fourth mora from the left edge of the verb stem.

Let us take this opportunity to situate Kuria in a broader Bantu context. While grammatical tones in Kuria target one or more of the first four TBUs of verb

Table 11: Kuria inceptive assigns grammatical tone into next word

to-ra-{karaaŋg-á éyétóókè	‘we are about to fry a banana’
to-ra-{sukur-a éyétóókè	‘we are about to rub a banana’
to-ra-{rom-a eyétóókè	‘we are about to bite a banana’
to-ra-{ry-a eyetóókè	‘we are about to eat a banana’

stem, additional targets are found in other Bantu languages. Example (1) summarizes the docking targets found in the more than two dozen languages as described in [Odden & Bickmore \(2014\)](#), a special volume of *Africana Linguistica*. (All references within these examples are from that volume – see that source for referential codes and geographic locations.)

- (1) Position in stem targeted by grammatical tones across Bantu
 - a. **1st:** Jita ([Downing 2014](#)), Kuria ([Marlo et al. 2014](#)), Nyala-West ([Ebarb et al. 2014](#)), Orungu ([Maniacky & Ambouroue 2014](#)), Simakonde ([Manus 2014](#)), Wanga ([Ebarb et al. 2014](#))
 - b. **2nd:** Cilungu ([Bickmore 2014](#)), Copperbelt Bemba ([Bickmore 2014](#)), Emakhuwa ([Kisseberth & Guérois 2014](#)), Fipa ([Bickmore 2014](#)), Idhakho ([Ebarb et al. 2014](#)), Kamba ([Roberts-Kohn 2014](#)), Kuria ([Marlo et al. 2014](#)), Namwanga ([Bickmore 2014](#)), Totela ([Crane 2014](#))
 - c. **3rd:** Emakhuwa ([Kisseberth & Guérois 2014](#)), Kuria ([Marlo et al. 2014](#))
 - d. **4th:** Kuria ([Marlo et al. 2014](#))
 - e. **Antepenult:** Bena ([Morrison 2014](#))
 - f. **Penult:** Bena ([Morrison 2014](#)), Emakhuwa ([Kisseberth & Guérois 2014](#)), Jita ([Downing 2014](#)), Kamba ([Roberts-Kohn 2014](#)), Simakonde ([Manus 2014](#))
 - g. **Final:** Basaa ([Makasso 2014](#)), Bakweri ([Marlo & Odden 2014](#)), Cilungu ([Bickmore 2014](#)), Copperbelt Bemba ([Bickmore 2014](#)), Emakhuwa, Fipa ([Bickmore 2014](#)), Jita ([Downing 2014](#)), Kamba ([Roberts-Kohn 2014](#)), Kwanyama ([Halme-Berneking 2014](#)), Mbadja ([Halme-Berneking 2014](#)), Namwanga ([Bickmore 2014](#)), Nyala-West ([Ebarb et al. 2014](#)), Orungu ([Maniacky & Ambouroue 2014](#)), Totela ([Crane 2014](#)), Wanga ([Ebarb et al. 2014](#))

In addition to those listed above, Nyala-West has a pattern which targets the first mora of the second stem syllable, and Wanga and Kifuliiru have patterns which target the leftmost free TBU.

That same volume details the range of tones and combinations of tone within each language (as analyzed by the authors) which are used as grammatical tone melodies. Example (2) summarizes this for the languages examined above. (Note that sL= super-low and sH= super-high).

(2) Grammatical tone melodies across Bantu

- a. **Just H:** Basaa (Makasso 2014), Bena (Morrison 2014), Emakhuwa (Kisseberth & Guérois 2014), Ecuwabo (Kisseberth & Guérois 2014), Jita (Downing 2014), Kuria (Marlo et al. 2014), Cilungu (Bickmore 2014), Idhako (Ebarb et al. 2014), Nyala (Ebarb et al. 2014), Wanga (Ebarb et al. 2014), Makwe (Devos 2014), Mbadja (Halme-Berneking 2014), Shona (Odden 2014), Totela (Crane 2014)
- b. **H, L:** Orungu (Maniacky & Ambouroue 2014)
- c. **H, LH:** Dawida (Philippson 2014)
- d. **H, L, LH:** Dibole (Leitch 2014)
- e. **H, L, LH, HL:** Bakweri (Marlo & Odden 2014)
- f. **H, L, sH, sL, H-sL:** Kamba (Roberts-Kohno 2014)
- g. **H, L, LH, HL, HH, LL:** Kifuliiru (Van Otterloo 2014)
- h. **LH, LHL:** Lulamogi (Hyman 2014)
- i. **H, L, LH, LHL:** Simakonde (Manus 2014)

Bantu languages thus differ in terms of their inventory of grammatical tones and where these grammatical tones associate. One of the languages found above, Orungu ([mye] – Gabon), shows that languages can also differ as to the behavior of grammatical tone akin to the divisions we made in Section 1.2 (e.g. additive, subtractive, replative, *etc.*). Most grammatical @ tones affect their local environment, but one sponsored by the negative prefix à- NEG is different: it is a replative pattern which spreads to all TBUs to its right. A remarkable aspect of this pattern is that its scope is unbounded: the grammatical tone replaces the underlying tone even of whole words to its right, until a clause boundary. A data set is provided in Example (3).

(3) Orungu replative grammatical tone (Ambouroue 2007, Maniacky & Ambouroue 2014)

- a. [à-yól-á]
à@-yòl-a
NEG-buy-FV
'don't buy!'

- b. [à-γómb-íz-ó-ní]
 àⓂ-γómb-íz-ò-nì
 NEG-chase-CAUS-PASS-PL
 ‘don’t be chased away!’
- c. [à-ríy’ áwán’ áṅká yó ’ṅkóló]
 àⓂ-rìy-à àwáná áṅká yó ìṅkòlò
 NEG-leave-FV children alone P tonight
 ‘do not leave the children alone tonight’

Bantu is far from the only sub-family in Benue-Congo which exhibits grammatical tone. In non-Bantu languages westward in Nigeria, such tonal effects are also found but often without the characteristic ‘flare’ of Bantu verb inflection. In Yoruba ([yor]), a sizable literature has emerged on the function of the so-called subject-marking ‘High Tone Syllable’ (HTS) which appears between the subject and predicate in many (if not most) inflectional environments (Awobuluyi 1975, Bisang & Sonaiya 1999, *inter alia*). The HTS is realized either as a copy of the preceding vowel (i.e. the last vowel of the subject) or it replaces the final tone of the subject. For example, /bābā/ ‘father’ becomes either [bābā á] or [bābá] in the first row of Table 12.

Table 12: Yoruba grammatical Ⓜ at subject-predicate boundary (Awobuluyi 1975)

bābā Ⓜ lẹ	→	[bābā á lẹ] ~ [bābá lẹ]	‘Father went’
wọn Ⓜ lẹ	→	[wọn ọn lẹ] ~ [wọn lẹ]	‘they went’
ògèdè Ⓜ wọn	→	[ògèdè ẹ wọn] ~ [ògèdè wọn]	‘bananas are expensive’

Moreover, in the fellow Nigerian language Igbo ([ibo]) a low tone appears at the subject-predicate boundary to indicate a simple polar question, i.e. ① Q. This grammatical tone associates to the prefix on the verb but then also spreads backwards onto any contiguous string of high tones on the subject, shown in example (4).

- (4) Igbo polar question ① subject-predicate boundary (Clark 1990)
- a. [hà àlá’yí]
 há ① á-’lá’yí
 they Q PRF-leave\NEG
 ‘didn’t they leave?’

- b. [ùwè ọchà ọ-dì n-ébè à]
 ùwè ọchà ① ọ-dì n-ébè à
 shirt clean Q PRF-is in-place this
 ‘Is there a clean shirt here?’

Grammatical tone patterns could also easily be shown for the other Benue-Congo families (e.g. Kainji, Edoid, Platoid, Nupoid, Delta Cross, Jukunoid, or non-Bantu Bantoid such as Grassfields).

2.2 Kwa

Another Niger-Congo family is Kwa, containing languages related to Akan/Twi and Ewe, among others. In the Gã language ([gaa] – Ghana), grammatical tones appear after the subject marker at the left edge of the verb. The Habitual sponsors a floating ① which docks to the preceding subject marker if it is toneless (all but the first row of Table 13) but results in downstep ¹ if the subject marker is high-toned (the first row).

Table 13: Gã floating tone from habitual (Paster 2003)

Subject	Habitual ‘X sing(s)’
1S mí-	mí-①-lá-ɔ → mí- ¹ lá-à
2S o-	o-①-lá-ɔ → ò-lá-à
3S e-	e-①-lá-ɔ → è-lá-à
1P wɔ-	wɔ-①-lá-ɔ → wɔ-lá-à
2P nyẽ-	nyẽ-①-lá-ɔ → nyẽ-lá-à
3P amẽ-	amẽ-①-lá-ɔ → àmẽ-lá-à

In contrast, inflection from the Subordinate grammatical ② and the Simple Past grammatical ③ are different: they always dock to the preceding subject marker and neutralize any underlying tone difference (the H vs. Ø distinction). These patterns are in Table 14. Observing the difference between the Habitual and the Simple Past specifically, this demonstrates that the value of the grammatical tone is not enough to predict behavior.

In another Kwa language, Saxwe ([sxw] – Benin), we find evidence of grammatical tones being prioritized for survival over tones with no meaning such as boundary tones. Clauses by default are realized with a meaningless L% boundary tone (BT) at the right edge of a clause (though a small number of contexts have a

Table 14: Gã neutralization of subject agreement by inner inflection

Subject	Subordinate ‘that X sing’			Simple past ‘X cultivated’		
1S má-/mí-	má-Ⓜ-lá	→	má-lá	mí-Ⓛ-dú	→	mí-dú
2S o-	o-Ⓜ-lá	→	ó-lá	o-Ⓛ-dú	→	ò-dú
3S e-	e-Ⓜ-lá	→	é-lá	e-Ⓛ-dú	→	è-dú
1P wɔ-	wɔ-Ⓜ-lá	→	wó-lá	wɔ-Ⓛ-dú	→	wò-dú
2P nyẽ-	nyẽ-Ⓜ-lá	→	nyé-lá	nyẽ-Ⓛ-dú	→	nyè-dú
3P amẽ-	amẽ-Ⓜ-lá	→	àmé-lá	amẽ-Ⓛ-dú	→	àmè-dú

H% boundary tone, e.g. certain clauses with a syntactic gap). We exemplify the surface forms when a L% boundary tone is present in Example (5), with both final high-toned and mid-toned words. We see that only the mid-toned verb /sẽ/ changes, becoming a falling mid-low tone [sẽ] – high tones remain unaltered.

(5) Saxwe clause-final boundary tone L% (Beavon-Ham 2019)

- a. [õxẽ s̃́]
õxẽ s̃́ L%
bird leave BT
‘a bird left’
- b. [õxẽ sẽ]
õxẽ sẽ L%
bird hear BT
‘a bird heard’

In contrast, there also exists a clause-final grammatical ① which is part of the expression of negation (along with the pre-verbal particle ð̃ NEG). Unlike with the boundary tone, with negative grammatical tone both underlying mid and high are affected, i.e. /s̃́/ becomes [s̃̃] (6a) just as /ñ̃/ becomes [ñ̃̃] (6b).

(6) Saxwe clause-final grammatical ① expressing negation

- a. [kõfí ð̃ s̃̃]
kõfí ð̃ s̃̃ ①
Kofi NEG leave NEG
‘Kofi did not leave’

- b. [ōsós ǝ gbè nǝ̃] [ǝ̃]
 ōsós ǝ ⑤ ⑥ gbě nǝ̃ ⑦
 horse NEG IRR IPFV refuse IPFV NEG
 ‘the horse will not refuse’

2.3 Kru

To the west of the Kwa family are the Kru languages (sometimes not even included in Niger-Congo, e.g. in Glottolog – [Hammarström et al. 2025](#)). Exemplary of grammatical tone in Kru is Guébie ([gie] – Côte d’Ivoire). Negation is expressed as a high grammatical tone ④ which appears between the subject and predicate (Guébie has four pitch heights which Sande transcribes numerically with 4, 3, 2, and 1, where 4 is the highest and 1 the lowest). Example (7) shows that the subject /ja³¹/ ‘coconuts’ with an underlying falling contour surfaces as [ja³¹⁴] with a dipping contour after the floating ④ docks in the negative.

(7) Guébie grammatical ④ marking negation on subjects ([Sande 2022](#))

- a. [ja³¹ na⁴nɛ⁴]
 ja³¹ na⁴nɛ⁴
 coconuts be.good
 ‘coconuts are good’
- b. [ja³¹⁴ na⁴nɛ⁴]
 ja³¹ ④ na⁴nɛ⁴
 coconuts NEG be.good
 ‘coconuts are not good’

A complete set of patterns are provided in Table 15, which demonstrates the straightforward addition of the negative tone regardless of which tone the subject ends in. This set also demonstrates the absence of neutralization, as even words with underlying /4/ tone show lengthening in the context of negative ④ (e.g. /ju⁴/ ‘water’ surfaces as [ju⁴u⁴]).

Even more intriguing is the marking in imperfective which shows a scalar shift. In imperfective contexts, the initial tone of the verb lowers one level in pitch, e.g. /li²ʔe³/ ‘dine’ (8a) surfaces as [li¹ʔe³] in imperfective contexts (8b). In the latter, we gloss this simply as a morpheme ⑩ IPFV, standing for the scalar tone effect.

(8) Guébie imperfective scalar shift ([Sande 2018](#))

Table 15: Guébie summary of grammatical ④ NEG (Sande 2022)

Underlying	Under negation
/...1/	→ ...14
/...2/	→ ...24
/...3/	→ ...34
/...4/	→ ...44
/...23/	→ ...234
/...24/	→ ...244
/...41/	→ ...414
/...42/	→ ...424
/...31/	→ ...314
/...32/	→ ...324

- a. [ɔ³ li²ʔe³]
 ɔ³ li²ʔe³
 3s.HUM dine
 ‘(s)he dined’
- b. [ɔ³ li¹ʔe³]
 ɔ³ ① li²ʔe³
 3s.HUM IPFV dine
 ‘(s)he is dining’

The Guébie scalar shift shows a unique property, as well: if the verb is underlying /1/ and is thus already at the bottom of the pitch scale, it forces the preceding word to raise, i.e. the final tone of the subject shifts upward, e.g. 3SG $\epsilon^3 \rightarrow \epsilon^4$ in Table 16. Remarkably, if the final subject’s tone is already at the top of the scale /4/ then a derived super-high tone surfaces [5] which is otherwise not used in the language.

Table 16: Guébie scalar shift affects subject with /1/ verbs (Sande 2018)

ϵ^3 ʔɔ ¹	→	ϵ^3 ʔɔ ¹	‘it withered’
ϵ^3 ① ʔɔ ¹	→	ϵ^4 ʔɔ ¹	‘it withers’ (Imperfective)
e^4 pa ¹	→	e^4 pa ¹	‘I ran’
e^4 ① pa ¹	→	e^5 pa ¹	‘I run’ (Imperfective)

2.4 Gur

To the general north of the Kwa and Kru area is the Gur family of Niger-Congo, where grammatical tone again is omnipresent. In one major Gur language, Dagbani ([dag] – Ghana), there exist the typical grammatical tone patterns found across Africa, such as marking noun compounds and verb inflection involve negation, tense, and aspect. Dagbani contrasts underlying low-toned roots (e.g. *pìi* ‘choose’) versus high-toned roots (e.g. *gbáá* ‘catch’), whose tones are altered in context. The Present Perfective replaces this distinction with a uniform ①(②) pattern (Table 17), while the negative equivalent replaces it with just ② (note orthogonal downstepping and spreading rules). In contrast, other contexts do not neutralize underlying verb tone. They either preserve the tone pattern on the verb itself (e.g. the Relative Present Perfective in the third row in Table 17), or effects of the underlying tone are felt on the surrounding environment (e.g. downstep vs. spreading on the object in the last row with the Future). Such data also show that whether inflection is at the matrix clause level or within an embedded clause can affect grammatical tone assignment, found here in Dagbani but more robustly throughout Bantu.

Table 17: Dagbani tone inflection on verbs, with object *kòdú* ‘banana’ (Hyman & Olawsky 2004)

Inflection	Low-toned roots	High-toned roots
Present	ò pìi kó'dú	ò gbàà kó'dú
perfective	‘he has chosen a banana’	‘he has caught a banana’
Negative pres.	ò bá píi kòdú	ò bá gbáá kòdú
perfective	‘he hasn’t chosen a banana’	‘he hasn’t caught a banana’
Relative pres.	ɲùn pìi kòdú	ɲùn gbáá kòdú
perfective	‘...who has chosen a banana’	‘...who has caught a banana’
Future	ò-n píi kó'dú	ò-n gbáá kòdú
	‘he will choose a banana’	‘he will catch a banana’

In addition to sensitivities to tense, aspect, negation, and clause-type, the linguistic category mood can also dictate grammatical tone patterns. This is seen in another Gur language Turka ([tuz] – Burkina Faso), where a grammatical ② docks onto the subject to express the subjunctive (sjrv). In this language, the ② only docks to the right edge of the subject and cannot extend to other syllables or words past this position.

- (9) Turka subjunctive ⑩ marking on subject (Monich 2020, citing Suggett 2005)
- a. [wú¹kú]
wù ⑩¹kú
3SG SJTV die\SJTV
‘he would die’ (Cf. indicative: [wù kù] ‘he dies’)
- b. [wù kúúgú¹cá]
wù kúúgù ⑩¹cá
3SG death SJTV difficult\SJTV
‘may his death be difficult’

2.5 Mande

The Mande family is only controversially included in Niger-Congo, but it exemplifies the same breadth of grammatical tone patterns as the core of the phylum. We can exemplify this with Mande languages located in different areas of the Mande spread zone: Vai ([vai] – Liberia), Soninke ([snk] – Mali), and Seenku ([sos] – Burkina Faso). For further overviews of Mande grammatical tone, see Green (2018), Konoshenko (2018, 2022), *inter alia*.

First, the Vai language of Liberia shows grammatical tone in both its nominal and verbal system (Welmers 1976). Nouns contrast for tone, such as the minimal triplet in Table 18 showing H vs. L vs. LH (rising tone). To mark the definite versions of these nouns, one adds an autonomous ①⑩ grammatical tone sequence to the right edge (thus, /làá/ ‘clay pot’ becomes [lǎǎ] ‘the clay pot’ with two rising tones).

Table 18: Vai grammatical tone marking definiteness on nouns (Welmers 1976)

H	láá	‘plankfish’	→	HR	lǎǎ	‘the plankfish’
L	làà	‘paddle’	→	LR	lǎǎ	‘the paddle’
LH	làá	‘clay pot’	→	RR	lǎǎ	‘the clay pot’

Moreover, on verbs one sees a replacive pattern in the imperative (IMP), where the underlying tone of the verb is replaced by a uniform low tone (10). One exception to this is the motion verb *táá* ‘go’ (10d) which maintains its underlying high tone in the imperative (it is ‘indomitable’ in the terms of Rolle 2018). Such patterns show that grammatical tone may be suspended on a case-by-case basis, sometimes word-by-word.

(10) Vai replacive grammatical tone marking imperative (Welmers 1976)

- a. [mú kùnè]
 mú ① kúné
 us IMP wake
 ‘wake us up!’
- b. [kǒŋ tiè]
 kǒŋ ① tié
 stick IMP cut
 ‘cut sticks!’
- c. [kǒŋě lèsì]
 kǒŋ-ě ① lési
 stick-DEF IMP catch
 ‘catch the stick!’
- d. [í táá] (Cf. *[í tàà])
 í Ø táá
 you IMP go
 ‘go!’

Moving northward, Soninke shows grammatical tone changes when multiple words are placed in a compound, where the first word is the modifier and the second element is the head (Creissels 2016), a common pattern in (Western) Mande. In such constructions, the first word loses all tone but its first toneme which then spreads to the end of that word (the second word – the head noun – is only minimally altered tonally). Underlying H-tone initial words of various shapes (e.g. H, HL, HHL, HLF) surface as all-high (the top four rows of Table 19), while L-initial words surface as all-low (the bottom five rows). One might view this as a grammatically-induced spreading operation, rather than the addition of any overt (floating) tone.

Finally, moving eastward into Burkina Faso, the Seenku language demonstrates a scalar shift akin to the Guébie one in (8), though much more reduced. To form plurals, in addition to suffixation, a noun’s super-low (sL) tones become low (L), high tones (H) become super-high (sH), and super-high tones remain as such (there are incidentally no underlying L-toned nouns in Seenku). This is demonstrated in Table 20 (as with the Guébie example, the circled ① stands for the tone process).

Table 19: Soninke grammatically-induced spreading operation of the first word in a compound (Creissels 2016)

Change	Underlying		Compound	Meaning
H→H	tááxú hô	→	tááxú-hô	‘seat’ (lit. sit-thing)
HL→H	yúgò séntààdê	→	yúgú-n-céntààdê	‘man’s comb’ (lit. man-comb)
HHL→H	líhíndì màsî	→	líhíndí-mási	‘sewing machine’ (lit. sew-machine)
HLF→H	bánbàrà-llènmê	→	bánbárá-llènmê	‘young Bambara’ (lit. Bambara-DIM)
L→L	sì hàtê	→	sù-hàtê	‘horse skin’
LH→L	sèhé-llénmà	→	sèhè-llénmà	‘interlocutor’ (lit. speak-DIM)
LF→L	kìidê táxàyê	→	kìidì-táxàyê	‘Baobab condiment’
LHL→L	tùbàábù qálàsì	→	tùbààbù-qálàsì	‘European money’
LLF→L	sòònìnkê qánnè	→	sòònìnkà-n-qánnè	‘Soninke language’

Table 20: Seenku scalar shift in plural formation (McPherson 2017)

Change	Example		Plural form	Meaning
sL→L	jũ-Ⓣ-i	→	jùì	‘hills’
H→sH	bí-Ⓣ-i	→	bǐ	‘goats’
sH→sH	sũ-Ⓣ-i	→	sǔi	‘antelopes’

3 Nilo-Saharan

Next, let us examine another Greenbergian phylum, Nilo-Saharan. Although this has not been proven to be a genealogical unit *per se* (e.g. Glottolog does not treat it as such – Hammarström et al. 2025), its various families all show robust grammatical tone patterns. We shall examine some of its most recognizable families, namely Nilotic, Central Sudanic, and Nubian. For space reasons, we do not discuss Saharan, Songhay, Maban, Fur, Ik, Surmic, or various isolates (e.g. Kunama).

3.1 Nilotic

Nilotic languages are known for complex tonal alternations on nouns, typically understood as a reflex of their grammatical case system. Consider the Nilotic language Tugen ([tuy] – Kenya). There are at least 15 distinct alternation patterns when comparing absolutive case (the elsewhere form of the noun) to its nominative case form. Representative nouns are provided in Table 21.

Table 21: Tugen grammatical tone expressing case (Jerono 2013)

Absolutive	Tone		Nominative	Tone	Meaning
kibùkàndiì	L	→	kibùkándíí	LH	‘guitar’
chèp-ò	L	→	chép-ò	HL	‘daughter of’
kwááríík	H	→	kwáàrìik	HL	‘tendon’
móitá	H	→	mòità	L	‘calf’
káályááng’ík	H	→	kààlyááng’ík	LH	‘flies’
mùrsiik	LH	→	mùrsiik	L	‘sour milk’
chèèmòòsíí	LH	→	chéémóósíí	H	‘ogre’
káámèè	HL	→	kààmèè	L	‘mother’
cháálwòók	HL	→	cháálwóók	H	‘wrongdoing’
séèsèè	HL	→	séèséé	HLH	‘dog’
láàkwéé	HLH	→	láákwéé	H	‘child’
òinóòsyék	HLH	→	òinòòsyèk	HL	‘rivers’
ùsíí	LHL	→	úsíí	H	‘thread’
bèèlyóóndé	LHLH	→	béèlyóóndé	H	‘elephant’
kíkóòmbéé	LHLH	→	kíkóòmbéé	HLH	‘cup’

Tone marking grammatical case is widespread within the Nilotic family of Northeast Africa, also including Maasai (Tucker & Mpaayei 1955), Datooga (Kießling 2007), Kipsigis (Toweett 1979, Kouneli & Nie 2021), and Jumjum (Andersen 2004). In this last language Jumjum ([jum] – Sudan), the underlying tone of nouns is replaced by ① in the so-called ‘antigenitive’ case (which marks the non-possessor noun in a possessive construction), illustrated in Table 22.

It is not just nouns but also verbs which can display complex tonal alternations. This is seen prominently in Dinka ([dib] – South Sudan). Here, grammatical tone may result in complex fusional patterns where derivational and inflectional meaning are melded together. In the Agar dialect (Andersen 1992), derivational categories expressing directionality (e.g. the Centripetal cp) and valency (e.g. the

Table 22: Jumjum antigenitive case marking (Andersen 2004)

Absolutive	Tone		Antigenitive	Tone	Meaning
ḍéɛŋ	H	→	ḍéɛŋ	L	‘cow’
yîm	HL	→	yîm	L	‘giraffe’
kùun	L	→	kùun	L	‘thorn’
ḍiir-gà	HH	→	ḍiir-gà	LL	‘birds’
cìcàm	HL	→	cìcàm	LL	‘knife’
càw-nà	LH	→	càw-nà	LL	‘arrow’
mín-nèní	HLH	→	mín-nèní	LLL	‘child’
bòróŋ-gù	LHH	→	bòróŋ-gù	LLL	‘clothes’

Benefactive BEN, the Antipassive AP) are interwoven with inflectional categories of finiteness, person, number, (non-)topicality, and voice (e.g. Passive PASS). Verbs contrast for tone underlyingly (H vs. L), and this can be preserved or replaced depending on the precise features being expressed. The tonal patterns of various inflected-derived cells are provided in Table 23, showing only the patterns on CVC roots. Surface patterns are always high (H), low (L), or falling (F) (there are other (supra)segmental on the root, not shown).

The grammatical tone patterns of Table 23 Dinka evidence two morphological ‘layers’, which following the Dinka tradition we call ‘In’ (for an ‘inner’ layer) and ‘Out’ (for an ‘outer’ layer). In the inner layer, the pattern is governed largely by derivational categories (at the top), as reflected by the uniform tone values in the columns. In contrast, in the outer layer the pattern is governed by inflectional categories (at the left), reflected by the uniform rows.

3.2 Central Sudanic

Another Nilo-Saharan family with robust grammatical tone is Central Sudanic. In Ngiti ([niy] – The Democratic Republic of the Congo), verbs are inflected both for tense/aspect/mood as well as subject agreement. In Table 24, the intransitive verb *ìkpè* ‘cough’ appears in various inflection contexts which systematically alter its tone (with or without additional affixation). The basic inflected form of the stem is provided in this table under ‘Basic’, which is the form of the verb without subject agreement morphology. Depending on the exact inflection, the stem bears various tonal patterns (viz. MH-L, HL-M, LL-L, or ML-M). This is the first inflectional layer.

Table 23: Dinka layering of inflection-derivation grammatical tone (Andersen 1992, Trommer 2011)

		H-toned CVC roots					L-toned CVC roots				
		Ø	BEN	CP	AP	BEN.AP	Ø	BEN	CP	AP	BEN.AP
In	FIN	L	H	L	F	F	L	F	L	F	F
	1SG/3SG	L	H	L	F	F	L	F	L	F	F
	PL	H	H	L	F	F	H	F	L	F	F
	NFIN	F	H	L	L	F	L	F	L	F	F
Out	NTOP.SBJ	H	H	H	H	H	H	H	H	H	H
	PASS.CT	F	F	F	F	F	F	F	F	F	F
	PASS	H	F	F	F	F	H	F	F	F	F
	2SG	L	L	H	L	L	L	L	H	L	L

To the right of these basic stems are forms which bear subject agreement prefixes, here with third singular *k-* 3SG ‘he/she’ which adds a grammatical ①. At this second layer, when grammatical tone is added it creates rising contour tones, maintaining the basic inflected stem’s tone when possible. (Note that this example places the subject agreement forms in the negative with an independent word *inzá* ~ *inzá* NEG – these forms better illustrate the relevant tonal patterns. The translations provided are approximate based on the description in Kutsch Lojenga (1994).)

Table 24: Ngiti layering of grammatical tone (Kutsch Lojenga 1994)

Inflection	Tone	Basic	With additional tone layer (NEG + <i>k</i> ①- 3SG)		
Past hab.	MH-L	ikpé-nà	<i>inzá</i> <i>k-i</i> kpé-nà	‘(s)he didn’t use to cough’	
Past cond.	HL-M	ikpè-nā	<i>inzá</i> <i>k-ikpè</i> -nā	‘if (s)he hadn’t coughed’	
Recent past	LL-L	ikpè-nà	<i>inzá</i> <i>k-ikpè</i> -nà	‘(s)he didn’t cough’	
Future	ML-M	ikpè-nā	<i>inzá</i> <i>k-i</i> kpè-nā	‘(s)he won’t cough’	

In the related Sara-Bongo-Bagirmi family, comparable inflectional tone shows a cross-linguistically rare pattern: sensitivity to the initial segment of the target. We can illustrate this with the Eastern Sara language Nar ([mwm] – Chad). Table 25 shows person/number affixes in an unmarked tense/aspect, translated as simple present. With consonant-initial roots (e.g. C-initial /tógō/ ‘wash’), the

tone of the root does not alter in these contexts. In contrast, vowel-initial roots show grammatical tone changes (e.g. V-initial /íǵĩ/ ‘sell’, with the same underlying tones). The most telling forms are the third person, which are prefixless and surface with underlying HM with C-initial roots (i.e. [tógō] in this table), but as LL bearing grammatical tone expressing subject agreement with V-initial roots (i.e. [ǵǵĩ]).

Table 25: Nar consonant-initial roots do not take grammatical tone but vowel-initial roots do (Keegan 2012)

	Tone	C-initial	Translation	Tone	V-initial	Translation
1S	M-HM	m̄-tógō	‘I wash’	HM	m-íǵĩ	‘I sell’
2S	M-HM	ĩ-tógō	‘you wash’	HM	íǵĩ	‘you sell’
3S	HM	tógō	‘(s)he washes’	LL	ǵǵĩ	‘(s)he sells’
1P	L-HM	j̄-tógō	‘we wash’	LM	j-ǵĩ	‘we sell’
2P	M-HM-M	ĩ-tógō-ĩ	‘you wash’	H-M	ǵ-ĩ	‘you sell’
3P	HM-M	tógō-n̄	‘they wash’	LL-M	ǵĩ-n̄	‘they sell’

Keegan shows that this distribution applies to monosyllabic roots as well (e.g. /ǵr/ ‘pull’ becomes [ǵr̄], but counterpart C-initial monosyllabic roots do not change), showing one cannot simply explain away these facts through prosodic footing or frozen morphology.

3.3 Nubian

While far smaller and less thoroughly documented than the Nilotic and Central Sudanic families, the Nubian family too shows grammatical tone, especially in marking nouns (albeit usually of the non-autonomous type whereby the tone co-occurs with overt affixation). In Nobiin ([fia] – Egypt), nouns in isolation bear underlying tone, typically a single high span at the right edge, or an all low pattern. To indicate plurality, these nouns are marked by a suffix, idiosyncratically selected by particular roots (e.g. -ìì, -ncìì, -ríí, and -kúú, among others). Regardless of which suffix is selected, in all contexts the underlying tone of the root is neutralized to low, shown in Table 26. This affects Nubian-derived vocabulary as well as that which is Arabic-derived (e.g. /áádèṁ/ → [àádèṁ-ríí] ‘people’).

Table 26: Nobiin neutralizing tone in plural (Werner 1987)

Underlying	Example	Plural form
H	ág	→ àg-ìì ‘mouths’
	éen	→ èèn-cìì ‘women’
	kám	→ kàm-ríí ‘camels’
	áy	→ ày-kúú ‘hearts’
L	gùd	→ gùd-ìì ‘grasses’
	tùù	→ tùù-ncìì ‘bellies’
	kòòg	→ kòòg-ríí ‘ravens’
	dir	→ dir-kúú ‘sandals’
HH	féntí	→ fènt-ìì ‘dates’
	úmbúú	→ ùmbùù-ncìì ‘palm stems’
	ájáw	→ àjàw-ríí ‘sweets’
	kúrsí	→ kùrsì-gúú ‘chairs’
LH	kàdíis	→ kàdìis-ìì ‘cats’
	àrrée	→ àrrèè-ncìì ‘waterfalls’
	mèrgír	→ mèrgìr-íí ‘belts’
	gàttí	→ gàttì-gúú ‘odors’
LHH	àgíssí	→ àgìss-ìì ‘fishes’
	shìgírtí	→ shìgìrtìì-ncìì ‘hairs’
	kòkóttí	→ kòkòttì-ríí ‘muds’
	bèltááyíb	→ bèltààyìb-úú ‘earrings’
LLH	èskàléé	→ èskàlèè-ncìì ‘waterwheels’
	gàngàréé	→ gàngàrèè-gúú ‘spikes’

4 Afro-Asiatic

Next consider the Afro-Asiatic phylum. While its most famous families are non-tonal (i.e. Semitic, Berber, and Egyptian), those branches which have tone show robust grammatical tone patterns. We shall discuss key examples in families Chadic, Cushitic, and Omotic (the last being most likely several independent families unrelated to the rest of Afro-Asiatic, but within its orbit).

4.1 Chadic

The most populous Chadic language is Hausa ([hau] – Nigeria), spoken by tens of millions, and this language shows rampant grammatical tone effects in its nomi-

nal and verbal system. Like the Nobiin facts in Table 26 above, plural formation systematically replaces the tone of the base noun. But unlike in Nobiin, the exact grammatical tone pattern varies according to the precise suffix. Table 27 shows that the plural suffixes *-óóCíí* (where ‘C’ is a copy-consonant of the root) and *-únàà* assign a replacive ⑥ tone to the root, while the suffix *-áí* assigns replacive ⑦.

Table 27: Hausa plural suffixes assigning either ⑥ or ⑦ replacive tone (Newman 1986, 2000, Inkelas 1998)

Replacive ⑥	gyàlè	→	gyál-óólíí	‘shawls
	óófíṣ	→	óófíṣ-óófíí	‘office’ (< English)
	kààsúwá	→	káásúy-óóyíí	‘markets’
	tààtsúúniyáá	→	táàtsúúniy-óóyíí	‘birds’
	hántàà	→	hánt-únàà	‘livers’
	rììgáá	→	rííg-únàà	‘gowns’
	àkààwú	→	ákááw-únàà	‘clerks’
	àgóógó	→	ágóóg-únàà	‘clocks’
Replacive ⑦	wàkíìlì	→	wàkìil-áí	‘representatives’
	ḑáálìbìí	→	ḑààlib-áí	‘student’
	àlmákàǰíí	→	àlmákàs-áí	‘scissors’ (< Arabic)
	mábùǰǰùgáá	→	màbùǰǰùg-áí	‘water springs’

Within the verbal system, replacive grammatical tone also plays a major role in derivational and inflectional morphology. One example is the imperative, which overwrites any lexical tone with a replacive ⑦⑥ pattern, an autonomous grammatical tone pattern. This is exemplified in Table 28.

Table 28: Hausa imperative expressed by replacive ⑦⑥ (Newman 2000, Trommer 2011)

LH	fitá	→	LH	fitá	‘go out!’
HL	tá:ǰì	→	LH	tà:ǰì	‘get up!’
HH	kwá:ná	→	LH	kwà:ná	‘spend the night!’
LHL	tàimákà:	→	LLH	tàimàkí	‘help!’
HLH	kàràntá:	→	LLH	kàràntá:	‘read!’
HHH	nánnè:mó:	→	LLH	nànnè:mó:	‘seek (them) repeatedly!’

At the same time, not all grammatical tone is replacive in Hausa (Inkelas 1998). One is the definite clitic = $\textcircled{n}$ $\sim$ = $\textcircled{r}$ ‘the’ which assigns an additive $\textcircled{n}$ tone to the target of grammatical tone (on the distribution of the two allomorphs, see Newman 2000). This specifically does not result in replacement of the entire word’s tones, but typically results in a falling contour.

Table 29: Hausa additive $\textcircled{n}$ sponsored by definite clitic (Newman 2000)

bàká:= $\textcircled{n}$	→	bàkân	‘the bow’
ùngùlú= $\textcircled{n}$	→	ùngùlûn	‘the vulture’
mánò:má:= $\textcircled{n}$	→	mánò:mân	‘the farmers’
gó:díyó:yí:= $\textcircled{n}$	→	gó:díyó:yîn	‘the mares’
mó:tà:= $\textcircled{r}$	→	mó:tàř	‘the car’

4.2 Cushitic

While less robustly tonal than Chadic, tonal Cushitic languages also show grammatical tone patterns, typically which comply with pre-existing aspects of their tonology (e.g. allowing only one H per word). One famous case is the correlation between the position of high tone and gender and number in Somali ([som] – Somalia), shown in Table 30. A general correlation in the language is that male and/or singular nouns have a H on the penultimate mora, while counterpart female and/or plural nouns shift the H to the final mora.

On these patterns in Somali (and also the Cushitic language Rendille), Mous (2021) cites Oomen (1981) on their origin via a former suffix, e.g. a (reconstructed) feminine suffix **-et* which shifted default high tone assignment and then was subsequently lost. See Mous as well for wider discussion of grammatical tone in Cushitic.

4.3 Omotic

Also generally included under the umbrella of Afro-Asiatic are the Omotic languages (split up in Glottolog into genealogically unrelated families North Omotic, South Omotic, Blue Nile Mao, and Dizoid – Hammarström et al. 2025). Despite their questionable status as Afro-Asiatic, they are still often described using terms from Afro-Asiatic and Semitic linguistics. For example, in Ganza ([gza] – Ethiopia) when roots are modified by certain morphological environments (e.g. by a possessor) they appear in a ‘construct state’ (e.g. Frajzyngier & Shay 2012:

Table 30: Somali marking gender/plural *via* tone shift (Hyman 1981)

Male and/or Singular		Female and/or Plural	
náil	‘male lamb’	naíl	‘female lamb’
ínan	‘son, boy’	inán	‘daughter, girl’
nácas	‘stupid man’	nacás	‘stupid woman’
qaálin	‘young male camel’	qaalín	‘young female camel’
daméer	‘he-donkey’	dameér	‘she-donkey’
caasáan	‘young he-goat’	ceesaán	‘young she-goat’
éi	‘dog’	eí	‘dogs’
túug	‘thief’	tuúg	‘thieves’
kálax	‘ladle’	kaláx	‘ladles’
bálli	‘reservoir’	ballí	‘resevoirs’
doofaar	‘pig’	doofaár	‘pigs’
Soomaáli	‘Somali man’	Soomaali	‘Somali people’

533 for some background in Afro-Asiatic). In Ganza, the changes from the underlying tones can be called ‘construct tone’. For example, in the construct state of lexical roots underlying tone is replaced by a construct tone pattern which is only partially phonologically predictable. Table 31 shows that construct tone reduces tonal contrast to three shapes: L, HL, or H¹ (note that the downstep ¹ and upstep ¹ symbols indicate changes on the following word).

For nouns and adjectives, construct tone is triggered by most modification. However, with verbs construct tone is triggered by what appears to be an idiosyncratic set of contexts. For example, the verbal clitic =*na* conditions the construct form of the verb /p’áp’í/ ‘gather’ (11a, where it surfaces with [L] construct tones), while the clitic =*bo* preserves the underlying high tones (11b). These two clitics are translated identically, and throughout Smolders’ work appear as the default inflectional forms conveying basic present/past tense (depending on the verb’s inherent semantics).

- (11) Ganza idiosyncrasy of construct tone triggers (e.g. =*na*) and non-triggers (e.g. =*bo*) (Smolders 2016)

- a. [hàgá p’àbìnà]
 hà’=ga p’àp’i=na
 AFF=3M.SBJ gather\CNST=CL
 ‘he gathers’

Table 31: Ganza underlying vs. grammatical ‘construct tone’ (Smolders 2016)

	Underlying			Construct		Meaning
Pattern 1	HH	tákú	→	LL	tàkù	‘run’ (PL)
	LL	kùrù	→	LL	kùrù	‘genet’
	HH ^l	t’úlá ^l	→	LL	t’ùlà	‘dikdik’
	LH	p’òfó	→	LL	p’òfò	‘chase’
	LH ^l	kjàlá ^l	→	LL	kjàlà	‘colobus’
Pattern 2	HL	tájà	→	HL	tájà	‘sojourn’
	H ^l H	kí’á	→	HL	kí’à	‘catch’
Pattern 3	HH ^l	sásó ^l	→	HH ^l	sásó ^l	‘monkey’
	LL ^l	wàṅà ^l	→	HH ^l	wáṅá ^l	‘chicken’
	H ^l H	ǰí’ní	→	HH ^l	ǰíní ^l	‘work’

- b. [hàgá p’ábíḃô]
hà^l=ga p’áp’í=bo
AFF=3M.SBJ gather=CL
‘he gathers’

5 Khoisan

Our last Greenbergian phylum is Khoisan, which at this point has been demonstrated as not forming a valid language family, but rather five (Güldemann & Fehn 2014): three families Khoe-Kwadi, Kx’a, Tuu, and two isolates Hadza and Sandawe. We provide only a brief discussion of Khoisan, leaving out entirely Kx’a, Tuu, and Hadza. Khoisan languages generally lack the grammatical tone flourish of other African family stocks in terms of nominal or verbal inflection (especially compared to nearby Bantu languages). Still, grammatical tone can be identified, most clearly with types of derivational morphology.

5.1 Khoe-Kwadi

The Khoe-Kwadi family contains the largest and perhaps best documented Khoisan language, Khoekhoe a.k.a. Khoekhoegowab or Nama/Damara ([naq] – Namibia). Its tone system has been extensively studied for over 100 years

(Haacke 1999, Haacke 2008, Brugman 2009, Kusmer 2020), and from this literature there is consensus that Khoekhoe has four pitch heights which combine to form six primary melodies distributed across a bimoraic lexical stem. A six-way minimal pair with an approximate citation pronunciation is provided in Table 32, to which we add an informal label for each melody and our working phonological analysis of the underlying tones (note 4=highest á, 3=high á, 2=low à, and 1=lowest à, and that post-vocalic /m/ is moraic and tone-bearing).

Table 32: Khoekhoe minimal pair of six melodies distributed across bimoraic stem (Haacke 2008)

Label	Approximate pitch	Underlying	Example	Meaning
‘Top’	[44] ~ [43]	/4/	!õńs	‘fist’
‘Higher-Rising’	[24] ~ [34]	/24/	!òńs	‘pollard’
‘Intermediate’	[33] ~ [32]	/3/	!óń	‘coagulate’
‘Level’	[22]	/2/	!òm	‘force out’
‘Bottom’	[11] ~ [12]	/1/	!õń	‘push’
‘Lower-Rising’	[12] ~ [13]	/12/	!òńs	‘udder’

Khoekhoe shows several ‘flip-flop’ rules (a term taken from the Chinese tone literature – e.g. Wang 1967), which appear in certain derivational contexts. One is weak flip-flop, which is triggered across numerous affixal and compound constructions. One productive trigger is the applicative suffix *-bǎ*, which adds an argument to the verb structure. An example set is provided in Table 33, which shows that the six underlying melodies form three pairs which neutralize their distinction: /4/ and /24/ both become 24, /3/ and /2/ become 2, and /1/ and /12/ become 12.

The other type is strong flip-flop. This appears only sporadically, but is systematically triggered by causative reduplication, whereby causation is expressed by two parts: a suffixed reduplicant which bears a fixed /21/ tone (the second copy) and a tone change on the root (the first copy). The strong flip-flop tone change involves the same pairs as its weak counterpart, but with tonal exchange rather than neutralization. In the first pair in Table 34, underlying /4/ becomes 24 while underlying /24/ becomes 4. Equally, /3/ becomes 2 but /2/ becomes 3, and /1/ becomes 12 but /12/ becomes 1.

What unifies these flip-flop rules is that both reference a /2/ tone. In weak flip-flop (Table 33), a grammatical ② is either added to the stem (positioned at whichever edge creates a rising contour) or it overwrites stem tone (as with the

Table 33: Khoekhoe weak flip-flop triggered by applicative -*bǎ* (Haacke 1999)

Pair	Tone change	Example	Applicative form
Pair 1	/4/ → 24	kǔrú ‘make’	→ kùrú-bǎ ‘make for’
	/24/ → 24	!nàrí ‘steal’	→ !nàrí-bǎ ‘steal for’
Pair 2	/3/ → 2	óá ‘return’	→ òà-bǎ ‘return for’
	/2/ → 2	sàrì ‘visit’	→ sàrì-bǎ ‘visit for’
Pair 3	/1/ → 12	!nǎrǐ ‘drive’	→ !nǎrǐ-bǎ ‘drive for’
	/12/ → 12	ùrì ‘jump’	→ ùrì-bǎ ‘jump for’

Table 34: Khoekhoe strong flip-flop (tonal exchange) triggered by causative reduplication (Brugman 2009)

Pair	Tone change	Example	Causative form
Pair 1	/4/ → 24	!nǒǎ ‘smile’	→ !nòǎ-!nòǎ ‘make smile’
	/24/ → 4	!nòǒ ‘measure’	→ !nǒǒ-!nòǒ ‘estimate’
Pair 2	/3/ → 2	sóǎs ‘shade’	→ sòm-sòm ‘make shady’
	/2/ → 3	kòn ‘move’	→ kón-kòn ‘move to and fro’
Pair 3	/1/ → 12	!nǎǎ ‘love’	→ !nǎǎ-!nǎǎ ‘inspire to love’
	/12/ → 1	!nùwù ‘short’	→ !nùwù-!nùwù ‘shorten’

‘Intermediate’ /3/ melody). This results in the neutralization of the three paired melodies. In contrast, in strong flip-flop (Table 34) grammatical ② only appears on stems which lack 2 underlying, but 2 is deleted if one is already there (hence, exchange). Comparable grammatical flip-flop patterns are found in related Khoe-Kwadi languages, e.g. G|ui (Nakagawa 2006).

5.2 Sandawe

Last but not least is the Sandawe language ([sad] – Tanzania), of unclear affiliation to other Khoisan languages (but perhaps related to Khoe-Kwadi). In the Sandawe literature, a process of ‘grammatically-conditioned tone lowering’ is described (e.g. Hunziker et al. 2008, Eaton 2010, *inter alia*), whereby the tones of the second word of certain noun phrases systematically lower. This is shown with various possessive constructions, compounds, and [NOUN ADJECTIVE] constructions in Example (12). In these data, the first word’s tones remain intact while the second’s lower.

Despite the name of this ‘tone-lowering’ operation, both Hunziker et al. (2008) and Eaton (2010) show that the underlying tone contrast of the target is still phonetically detectable, and therefore not completely neutralizing. If the second word ends in an underlying low tone – whether simply low (12a) or a falling contour (12b–12c) – then the final syllable bears a super-low tone (more accurately, a low-to-super-low contour where the pitch falls to the lowest point in a speaker’s pitch range). In contrast, if the second word ends in high underlyingly (whether a contour or not), then the final syllable lowers less, only to a flat level-low pitch (12d–12f).

(12) Sandawe grammatically conditioned tone-lowering, but without complete neutralization (Hunziker et al. 2008, Eaton 2010)

- a. [hèwé hùmbù]
 hèwé hùmbù
 he cow
 ‘his cow’
- b. [ʰàtʰú k’ànkárà]
 ʰàtʰú k’ànkárà
 lion black
 ‘black lion’

- c. [kǒ:ŋgó: bǒ:k'ò]
 kǒ:ŋgó: bǒ:k'ò
 Gkoongoo spoon
 'Gkoongoo's spoon'
- d. [gélé màtò]
 gélé mátó
 Gele gourd
 'Gele's gourd'
- e. [lʰàtʃʰú ts'wǎ:]
 lʰàtʃʰú ts'wǎ:
 lion tail
 'a lion's tail'
- f. [hàpú k'àmè]
 hàpú k'àmé
 you beer
 'your beer'

The maintenance of this contrast may point to the role of phonological phrasing in accounting for these tone-lowering effects. Interestingly, a similar process is found in tone sandhi in Khoekhoe, whereby non-initial words change to a special 'sandhi' form (separate and distinct from the flip-flop rules of Tables 33 and 34 above). See Kusmer (2020) for details.

6 Conclusion

In this chapter, we examined grammatical tone data from approximately thirty different African languages from all over the continent, and made reference to properties of several dozen others. We defined and explored the differences between lexical and grammatical tone and ultimately showed that they are both robustly attested in languages of every phylum. In terms of the range of grammatical properties which can be conveyed using grammatical tone, we suggest that any inflectional or derivational property which can be expressed by segmental morphemes can also be expressed by tone. We showed that in some cases a grammatical tone on its own might signal the grammatical property being conveyed, while in other cases it might be one of several co-exponents of the grammatical expression. We observed that the way grammatical tone is actually implemented takes a variety of forms (e.g. addition, subtraction, replacement, shifting,

exchanging, and spreading) and these were all exemplified in various African languages.

Because of its suprasegmental nature, tone can be quite challenging to describe and accurately document. This is true of lexical tone, and even more so with regard to grammatical tone. But given the myriad grammatical properties that it expresses in whole or part, as illustrated in the sections above, its documentation and description is absolutely indispensable for a full understanding of such data. Careful and intense research on African tone languages has contributed substantially to our general understanding of grammatical tone, and we expect ongoing such research will deepen this even further.

Symbols and abbreviations

∅	toneless	ISO	International Organization for Standardization
Ⓣ	floating/grammatical tone	L	low tone
%	boundary tone	M	mid tone
↓	downstep	sH	super-high tone
↑	upstep	sL	super-low tone
C	consonant	TBU	tone-bearing unit
F	falling tone	V	vowel
H	high tone		
HTS	High Tone Syllable (Yoruba)		

Glosses

1	first person	HUM	human
2	second person	HYP	hypothetical
3	third person	IMP	imperative
AFF	affirmative	IPFV	imperfective
AP	antipassive	IRR	irrealis
BEN	benefactive	M	male
BT	boundary tone	NTOP	non-topic
CAUS	causative	NEG	negative
CL	clitic	P	preposition
CNST	construct	PASS	passive
CONS	consecutive	PL	plural
CP	centripetal	PRF	perfect
CT	circumstantial	Q	question
DEF	definite	RPST	remote past
DIM	diminutive	SBJ	subject
FUT	future	SG	singular
FV	final vowel (Bantu)	SJTV	subjunctive

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